Dairy food intake in relation to semen quality and reproductive hormone levels among physically active young men

STUDY QUESTION Is increased consumption of dairy foods associated with lower semen quality? SUMMARY ANSWER We found that intake of full-fat dairy was inversely related to sperm motility and morphology. These associations were driven primarily by intake of cheese and were independent of overall dietary patterns. WHAT IS KNOWN ALREADY It has been suggested that environmental estrogens could be responsible for the putative secular decline in sperm counts. Dairy foods contain large amounts of estrogens. While some studies have suggested dairy as a possible contributing factor for decreased semen quality, this finding has not been consistent across studies. STUDY DESIGN, SIZE, DURATION The Rochester Young Men's Study (n = 189) was a cross-sectional study conducted between 2009 and 2010 at the University of Rochester. PARTICIPANTS/MATERIALS, SETTING, METHODS Men aged 18–22 years were included in this analysis. Diet was assessed via food frequency questionnaire. Linear regression was used to analyze the relation between dairy intake and conventional semen quality parameters (total sperm count, sperm concentration, progressive motility, morphology and ejaculate volume) adjusting for age, abstinence time, race, smoking status, body mass index, recruitment period, moderate-to-intense exercise, TV watching and total calorie intake. MAIN RESULTS AND THE ROLE OF CHANCE Total dairy food intake was inversely related to sperm morphology (P-trend = 0.004). This association was mostly driven by intake of full-fat dairy foods. The adjusted difference (95% confidence interval) in normal sperm morphology percent was −3.2% (−4.5 to −1.8) between men in the upper half and those in the lower half of full-fat dairy intake ($P < 0.0001$), while the equivalent contrast for low-fat dairy intake was less pronounced [−1.3% (−2.7 to −0.07; P= 0.06)]. Full-fat dairy intake was also associated with significantly lower percent progressively motile sperm (P= 0.05). LIMITATIONS, REASONS FOR CAUTION As it was a cross-sectional study, causal inference is limited. WIDER IMPLICATIONS OF THE FINDINGS Further research is needed to prove a causal link between a high consumption of full-fat dairy foods and detrimental effects on semen quality. If verified our findings would mean that intake of full-fat dairy foods should be considered in attempts to explain secular trends in semen quality and that men trying to have children should restrict their intake. STUDY FUNDING/COMPETING INTEREST(S) European Union Seventh Framework Program (Environment), ‘Developmental Effects of Environment on Reproductive Health’ (DEER) grant 212844. Grant P30 {"type":"entrez-nucleotide","attrs":{"text":" DK046200","term_id":"187635970","term_text":"DK046200"}}DK046200 and Ruth L. Kirschstein National Research Service Award T32 DK007703-16 from the National Institutes of Health. None of the authors has any conflicts of interest to declare.